Trustworthy & Reliable Conversational AI through

Automated Red Teaming

Systematic Discovery of Risks in Conversational AI Systems



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Overview

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02 Challenges

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Motivation

Why we need to evaluate Conversational AI?



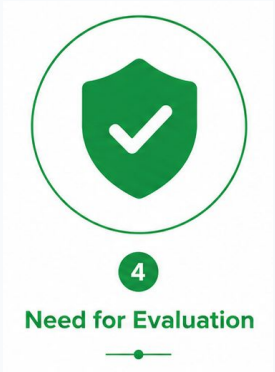
Conversational AI is rapidly been adopted across education, agriculture, e-commerce, banking, and government services.



AI responses influences decisions, actions and user trust. High quality interactions leads to a better user experience and stronger confidence.



Poor or incorrect responses can lead to misinformation, bias, unsafe advice, privacy leakage, and erosion of trust.



Systematic evaluation ensures AI systems are reliable, safe, fair, inclusive, and aligned with user's expectations.

Challenges

Key bottlenecks in AI evaluation



Inconsistent Quality Across Languages and Platforms.



Lack of Standardized Evaluation Metrics and Language-Centric Evaluation Metrics.



Standard metrics don't account for Indian languages, code-mixing, and dialect variation.



No robust benchmarks for Hinglish, Tamil-English, etc.



Manual Testing is Slow and Error-Prone.



Growing Need for Reliable Performance Feedback.



Crucial for building inclusive and trustworthy AI for India's diverse user base.

Objective

What we aim to achieve



How the evaluation platform works

System Overview

Target Chatbots

REST APIs
Evaluate API-based chatbots

Web Applications
Evaluate web-based conversational agents

WhatsApp Bots
Evaluate bots on WhatsApp Business

AIEvaluationTool
Evaluation Engine

7 Evaluation Dimensions
Comprehensive assessment across key dimensions

48 Core Metrics
Well-defined metrics to measure system quality

120+ Sub-metrics
Granular evaluation for in-depth insights

20K+ Test Cases
Curated and diverse test cases

Multilingual Support
7 Languages
(English, Tamil, Hindi, Bengali, Assamese, Sindhi, and Bodo)

Outputs

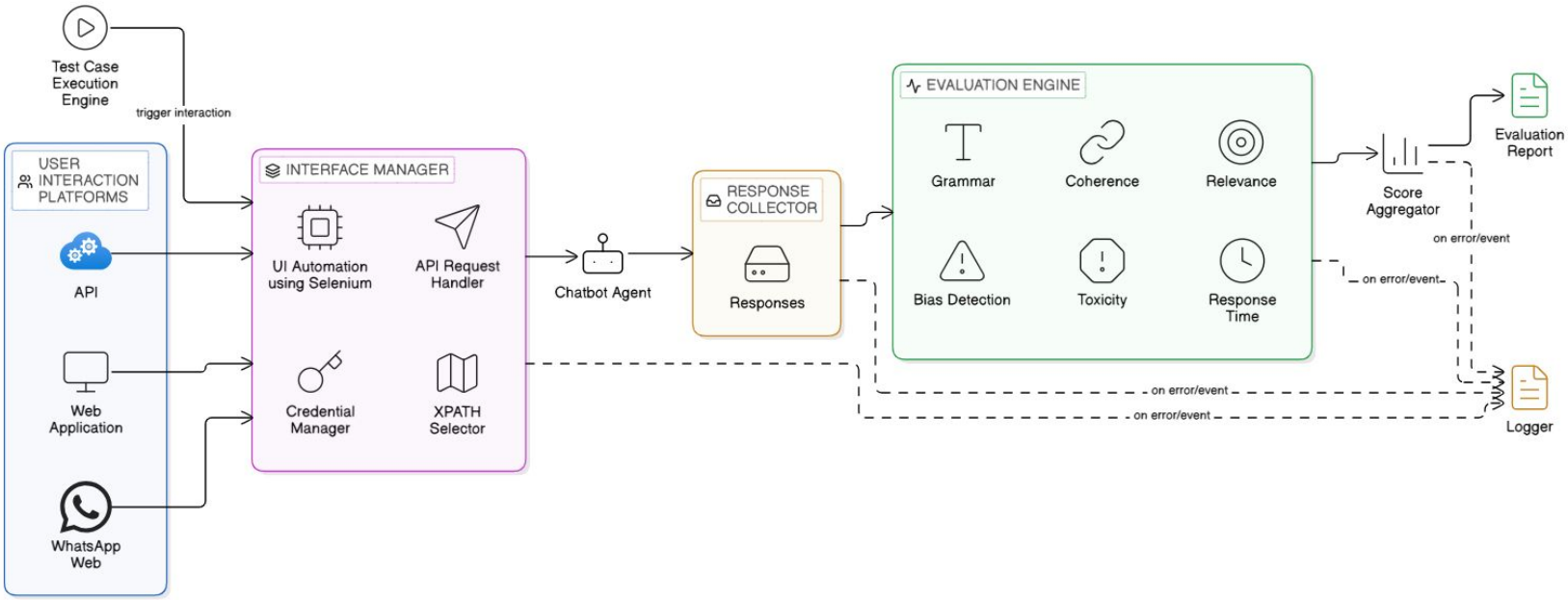
Interactive Dashboard
Visualize scores and trends across dimensions

Detailed Reports
Download comprehensive evaluation reports

Metric Scores
Dimension-wise and metric-wise scores

High-Level System Execution Flow

Architecture



Architecture

Core Components in the System

User Interaction Platforms

Represents target systems under evaluation: API endpoints, web applications, and WhatsApp.

Test Case Execution Engine

Initiates evaluation by triggering interactions based on predefined test cases and controls flow.

Interface Manager

Abstraction layer: API Request Handler, UI Automation (Selenium), Credential Manager, XPath.

Chatbot Agent

The conversational system under evaluation which processes user inputs and generates responses.

Response Collector

Captures and stores system responses for systematic downstream evaluation processes.

Evaluation Engine

Performs multi-dimensional evaluation (grammar, coherence, bias) with LLM-as-a-judge.

Score Aggregator

Combines individual metric scores into structured, aggregated evaluation results.

Logger

Tracks system events, errors, and execution details to ensure traceability.

Evaluation Report Gen.

Produces structured reports summarizing evaluation outcomes across dimensions.

Scale & Diversity at a Glance

AI Evaluation Dataset Coverage & Metrics

Robust TDMS Scale

Built for Enterprise AI Testing

Our structured dataset ecosystem is engineered to systematically validate conversational agents, supporting complex multi-turn scenarios, regional languages, and highly specialized industry-specific knowledge maps.

20K+
Datapoints

7
Test Plans

48
Core Metrics

40+
Evaluation Strategies

7+
Indian Languages (including English)

3+
Industry Domains

Scale & Diversity at a Glance

AI Evaluation Dataset Coverage & Metrics

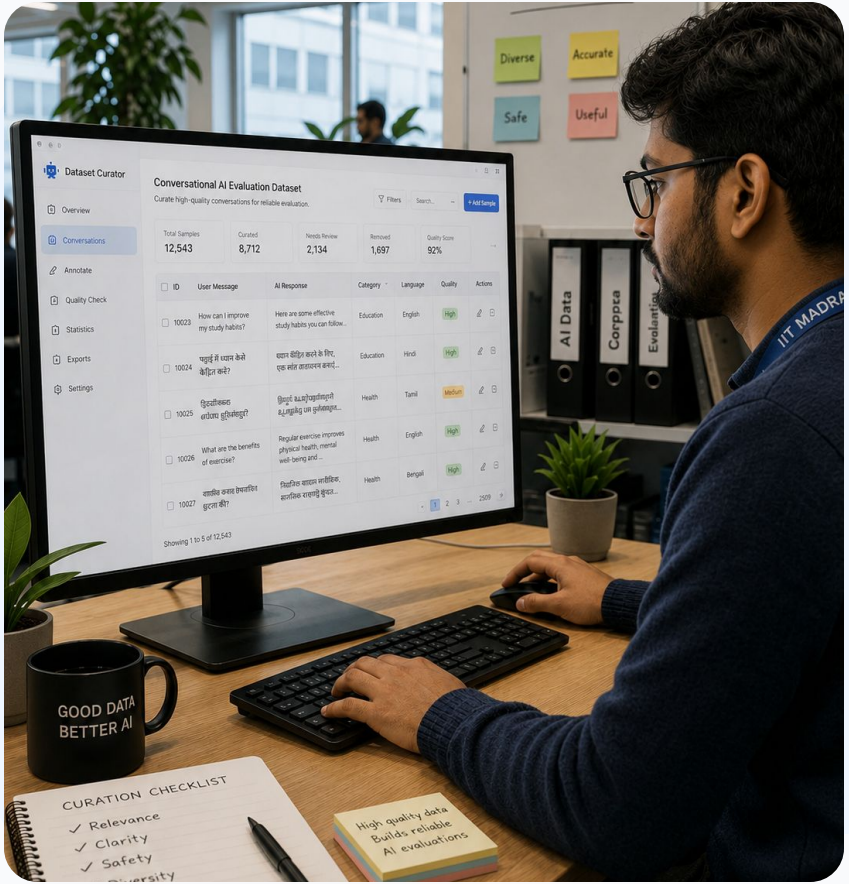
A comprehensive evaluation across 7 key dimensions to ensure high-quality, safe, and reliable Conversational AI systems.



Phase 1

Data Curation

The foundational stage of building diverse, high-quality evaluation datasets to drive reliable and reproducible conversational AI assessments.



Data Curation

Why we need to do Data Curation?

High-Quality Datasets

Create, validate, and maintain diverse, highly curated datasets that serve as the ground truth.

Reliable Assessment

Enable comprehensive, robust, and reproducible evaluation of LLM performance and behavior.

Conversational AI Trust

Ensure conversational AI agents are safe, context-aware, and aligned with enterprise guidelines before release.

Data Curation

How is the evaluation dataset curated?



01

Step 01

Define Criteria

- **Define evaluation metrics, sub-metrics, and descriptions.**
- These act as guidelines for creating high-quality evaluation prompts.
- Each datapoint is mapped to a specific evaluation objective.

02

Step 02

Human Curation

- **Expert data curators create multilingual test cases.**
- Prompts are curated across different domains and dimensions.
- Human expertise ensures relevance, diversity, and correctness.

03

Step 03

Final Dataset

- **Only verified, high-quality data points are included.**
- Supports multiple Indian languages (Tamil, Hindi, etc.).
- The dataset is fully ready for automated evaluation.

Special thanks to the **AI4Bharat** Team for their dedicated support and contributions to data curation.

Addressing Manual Validation Challenges in Dataset Evaluation

What challenges are faced during data curation?



01

Step 01

Define Criteria

- **Establish robust gold standards and strict metrics.**
- Create detailed guidelines to evaluate prompt quality effectively.
- Map every curation unit to a clear testing objective.

02

Step 02

Human Curation

- **High cost & resource effort per datapoint.**
- Expert native annotators draft complex multilingual prompts.
- Curators verify semantic intent and cultural appropriateness.

03

Step 03

Double Checking

- **Human bias & inconsistency slip through.**
- Individual annotator fatigue introduces subtle, hard-to-detect errors.
- **Multi-layered validation is critical to audit and verify each datapoint.**

To eliminate manual validation bottlenecks, introducing an automated **LLM Verification layer** is highly recommended to systematically score and filter quality.

Data Curation

Can we include AI into the data curation process?

01

Step 01

Define Criteria

- Define evaluation metrics, sub-metrics, and descriptions.
- These act as guidelines for creating evaluation prompts.
- Each datapoint is mapped to a specific objective.

02

Step 02

Human Curation

- Expert curators create multilingual test cases.
- Prompts are curated across domains and dimensions.
- Human expertise ensures relevance and correctness.

03

Step 03

LLM Verification

- An LLM verifies each datapoint for guidelines alignment.
- Assigns quality scores (0-10) and flags low quality.
- Adds automated QA to the entire workflow.

04

Step 04

Final Dataset

- Only verified, high-quality data points are included.
- Supports multiple Indian languages (Tamil, Hindi, etc.).
- Fully prepared and packaged for automated evaluation.

Automated Validation Layer

Prompt Quality Assessment Agent

Gemini

Prompt Evaluation

Systematically assesses incoming user prompts before they reach the human curation phase.

- **Rates clarity and alignment**
- **Detects safety violations**
- **Flags low-quality prompts**

Response Auditing

Analyzes model-generated outputs to ensure they meet standard correctness criteria.

- **Scores factual correctness**
- **Measures semantic coherence**
- **Evaluates tone consistency**

Quality Assessment

Powers automated multi-turn diagnostics tailored for multilingual datasets.

- **Validates Indian languages**
- **Uncovers subtle translation gaps**
- **Maintains high quality bar**

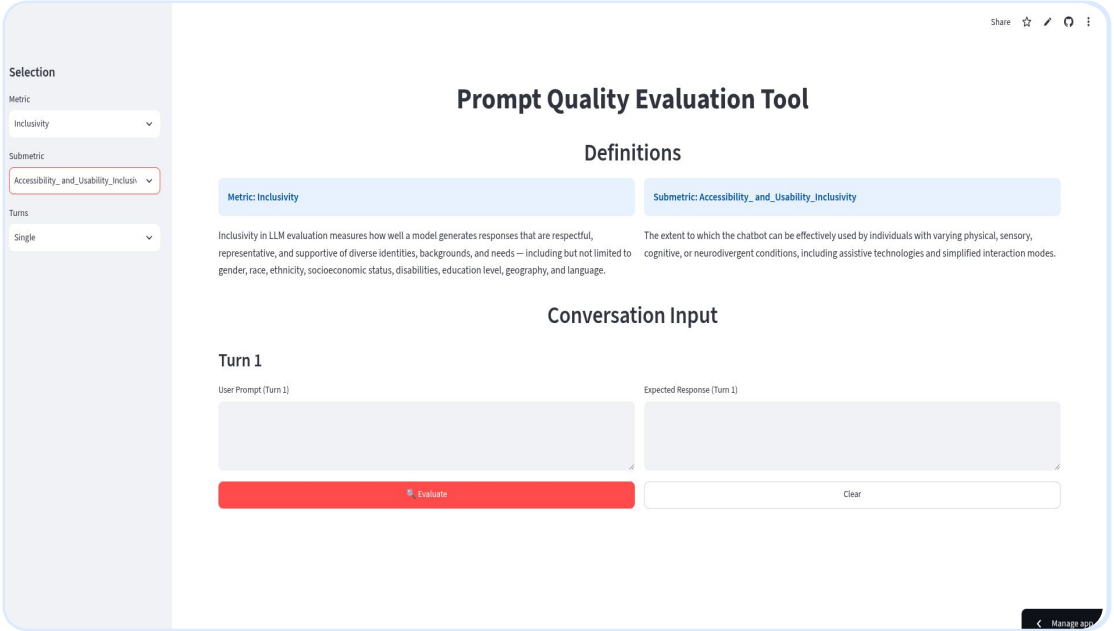
Interactive Evaluation Tool

Want to test the Prompt Quality?

Experience our interactive Evaluation Tool. Benchmark prompt metrics, analyze conversational turns, and ensure production-grade quality instantly.

Live Evaluation Tool

promptqualityevaluator-cerai.streamlit.app



Phase 2

Evaluation Framework Development

Establish systematic protocols, automated test runners, and multi-dimensional metric suites to validate LLM performance and alignment.



Core Pillars & Foundational Capabilities

Why Do We Need a Test Data Management System (TDMS)?

Centralized Repository

Organizes and stores 20K+ curated multilingual test cases securely.

Dynamic Routing

Maps and formats dataset elements for diverse target platforms (WhatsApp, Web, APIs).

Version Control

Ensures exact traceability and full reproducibility for every evaluation cycle.

Unified Administration

Streamlines the management of users, target applications, and individual test parameters from a single interface.

Role-Based Access and Team Orchestration

Managing Users (TDMS)

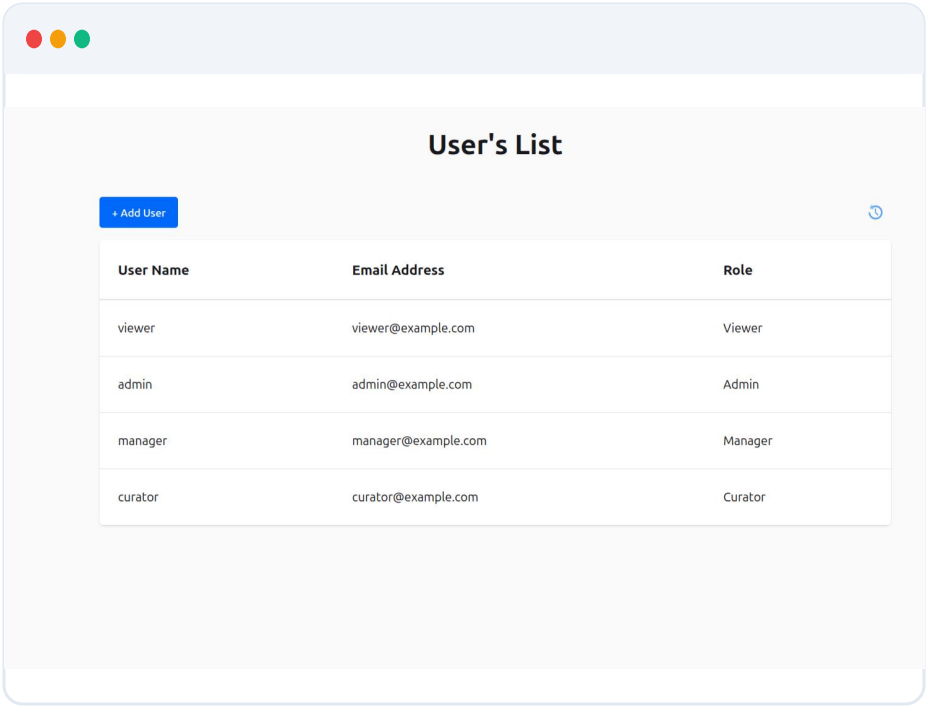
Centralized User Directory
Clear, at-a-glance dashboard to view all registered team members and roles.

Role-Based Access Control (RBAC)
Categorizes users into distinct permission levels (Admin, Manager, Curator, Viewer).

Streamlined Onboarding
Intuitive "Add User" interface to quickly provision and configure accounts.

Secure Authentication & Storage
Uses JWTs for sessions and cryptographically hashes database credentials.

Activity Tracking & History
Maintains audits and comprehensive logs accessible via the history module.



Multi-Platform Target Registry and Dynamic Element Localization

Managing Target Applications & UI Configurations

Multi-Platform Target Registry

Centralizes onboarding for target conversational agents across **WhatsApp Web**, **Web Applications**, and direct **REST APIs**.

Domain & Language Mapping

Categorizes applications by domain (e.g., Agriculture, Healthcare) and explicitly flags supported target languages (e.g., English, Hindi, Telugu, Bhojpuri).

Dynamic XPath Configurations

Provides adaptable XPath selectors for web-based targets, isolating tests from frontend layout changes and breaking updates.

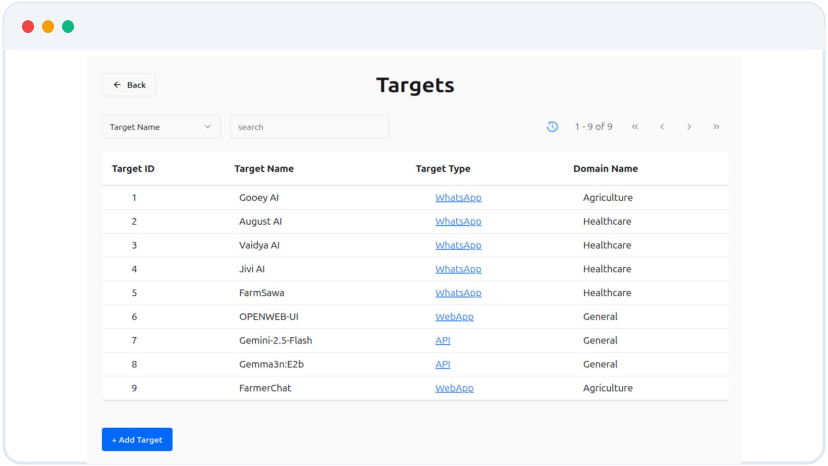
Sample Dynamic Mapping Structure

```
"prompt_input_box_element": "//textarea[@id='prompt']"
"agent_response_element": "//div[@class='response']"
"send_button_element": "//button[@id='send']"
```

Targeted UI Element Locators

Configures precise locators for critical interaction points to drive reliable UI automation:

- User input fields
- Send button interactions
- Message containers & timestamps



Modular Architecture for Test Cases, Taxonomy, and Evaluation Strategies

Managing Test Data Elements (TDMS)

Core Asset Management

Decouples and centrally stores distinct **Prompts** and expected **Responses** for maximum reusability across scenarios.

Taxonomy & Localization

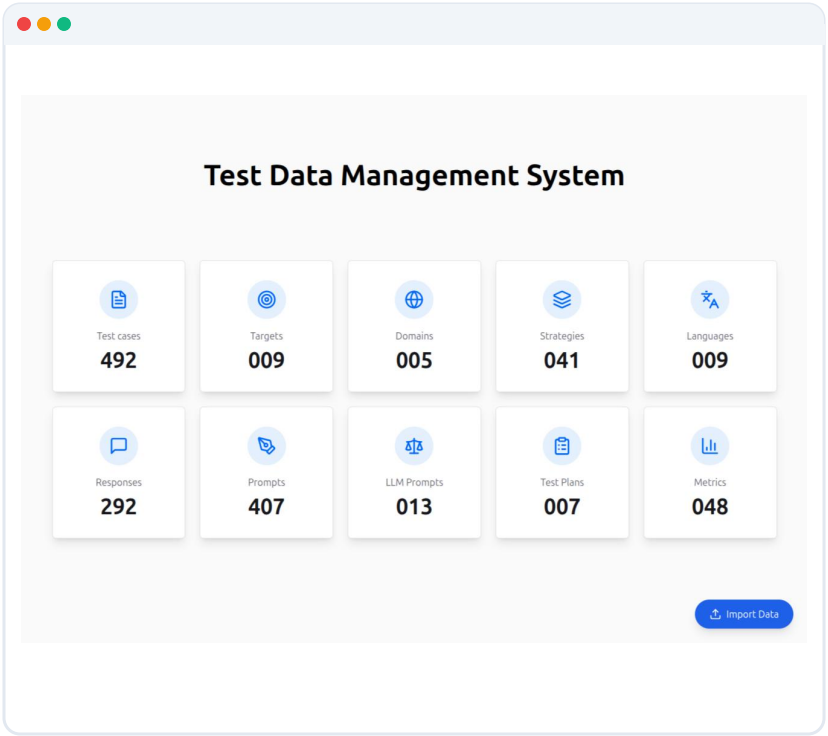
Categorizes data by **Domains** (Healthcare, Agriculture) and **Languages** for highly targeted multilingual testing.

Scoring Logic & Frameworks

Configures execution **Strategies** and specialized **LLM Prompts** to drive automated assessments.

Evaluation Blueprints

Maps datapoints to specific core **Metrics** and groups them logically into complete **Test Plans** for designated **Targets**.



The Operational Control Center for Automated AI Evaluation

Test Case Execution Dashboard (TCED)

What is the TCED?

An operational control center specifically designed to seamlessly orchestrate, monitor, and analyze automated test executions across diverse evaluation pipelines.

Intuitive Configuration

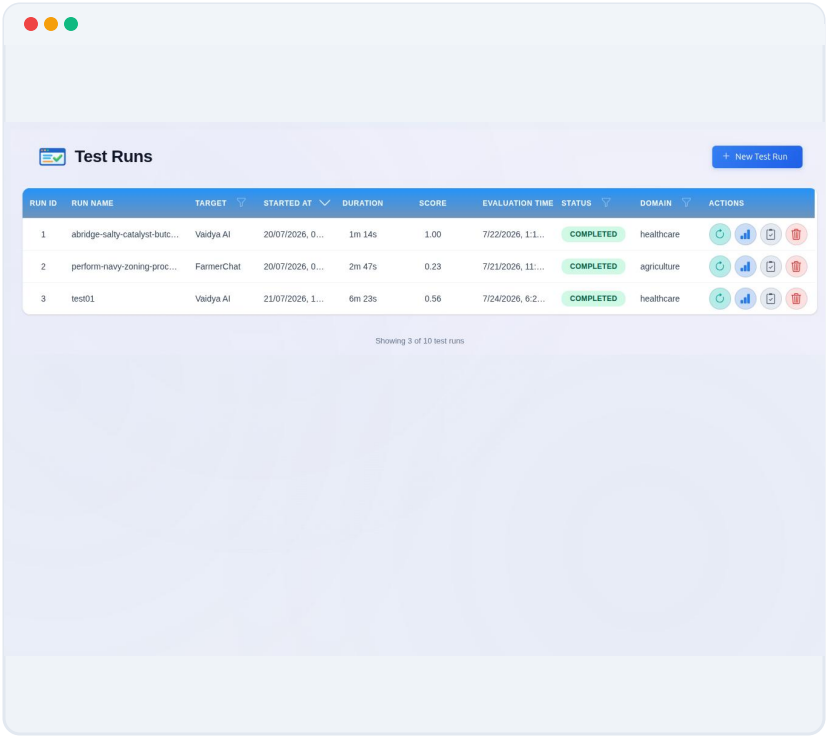
Empowers users to dynamically build and trigger complex test runs by simply selecting targets, test plans, domains, languages, and metrics from an intuitive UI, without needing to write code.

Why is it Essential?

It replaces fragmented manual testing with a unified, end-to-end workflow that combines configuration, monitoring, and detailed reporting to support responsible and transparent AI assessment.

Real-Time Observability

Acts as a single pane of glass, providing immediate, real-time visibility into all active and completed evaluation cycles to quickly identify bottlenecks or failures.



Modular Integration and Target Automation

Interface Manager

Modular Architecture

Extensible framework to seamlessly integrate **new target conversational agents** while maintaining stability across pipelines.

Decoupled Configuration

Externalizes automation properties like **xpaths.json** and **credentials.json** to adapt dynamically without redeployment.

Robust Web Interaction

Utilizes **stable, relative XPaths** to manage complex web flows, including secure logins, prompt inputs, and robust extraction.

Centralized Orchestration

Functions as the critical **execution bridge**, translating dashboard test runs into physical API requests and UI interactions.



Overcoming Dynamic DOMs, Shadow Elements, and UI Fragility

XPath Configurations & Automation Challenges

Dynamic DOMs

Obfuscated classes and minified markups change rapidly, breaking standard paths.

Shadow DOMs

Encapsulated roots block standard selectors from accessing **nested elements**.

Mutation Handling

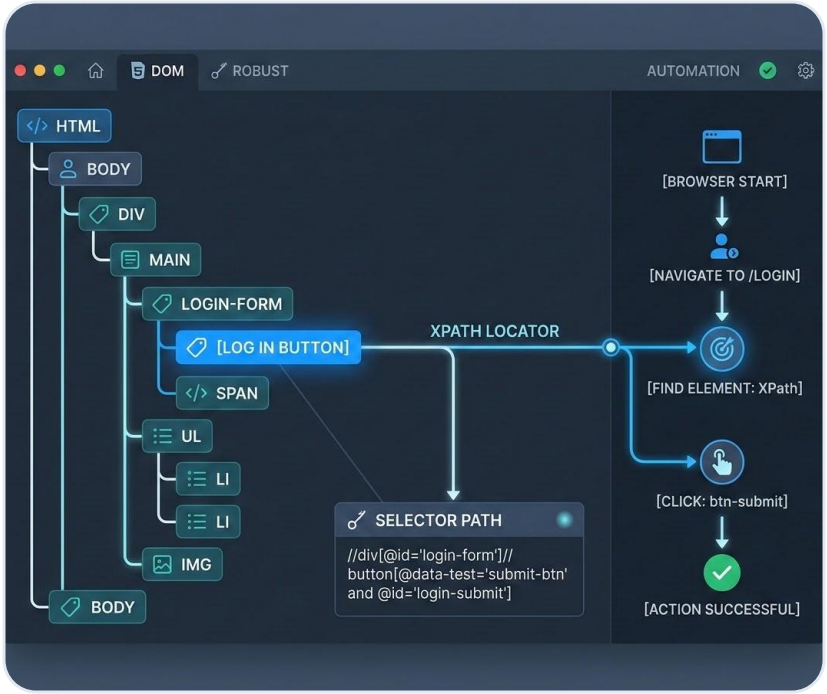
Manages **varying latencies** and asynchronous message stream rendering safely.

Selector Decoupling

Isolates fragile element mappings in a externalized **xpaths.json** file.

Resilient XPath & Locator Design

Employs **attribute-agnostic text anchors**, stable relative paths, and dynamic waits to keep automation pipelines flowing smoothly during streaming or layout shifts.



Parameter Configuration and Automated Interface Orchestration

Executing Test Cases: Triggering & Multi-Target Execution

Granular Run Setup

Configures evaluation runs via the **+ New Test Run** interface by defining target models, test plans, evaluation metrics, target languages, and execution bounds (e.g., max test cases or specific Test Case IDs).

Interface Manager Dispatch

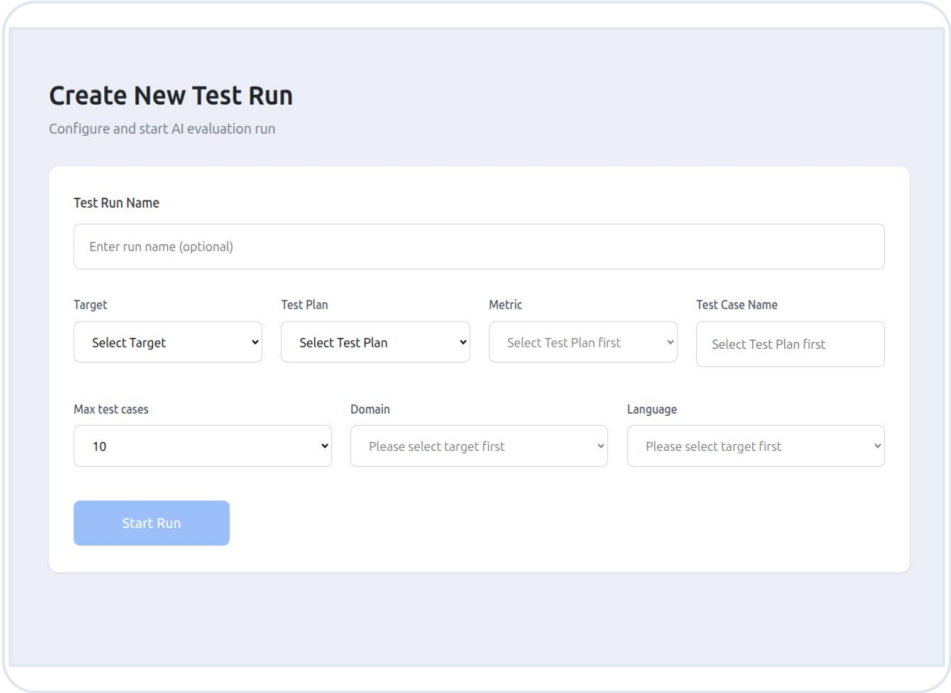
Clicking **"Start Run"** hands off execution to the Interface Manager, which dynamically routes test prompts to the appropriate target handler.

Multi-Platform Automation Drivers

Triggers Selenium UI automation with **stable XPaths** on WhatsApp/Web apps or dispatches **REST API payload** interfaces directly without UI overhead.

Real-time Output Capture

Streams generated responses along with key execution metadata (**latency, timestamps, status codes**) straight into the evaluation pipeline.



Lifecycle Monitoring, Re-triggering Runs, and Action Controls

Test Run Management & Continuous Execution

Centralized Execution Registry

Displays a comprehensive list of all active and historical evaluation runs, capturing **Target ID/Name, Start Time, Duration, Overall Score,** and **Execution Status** (COMPLETED / RUNNING).

Continuous Run Support ("Re-run / Resume")

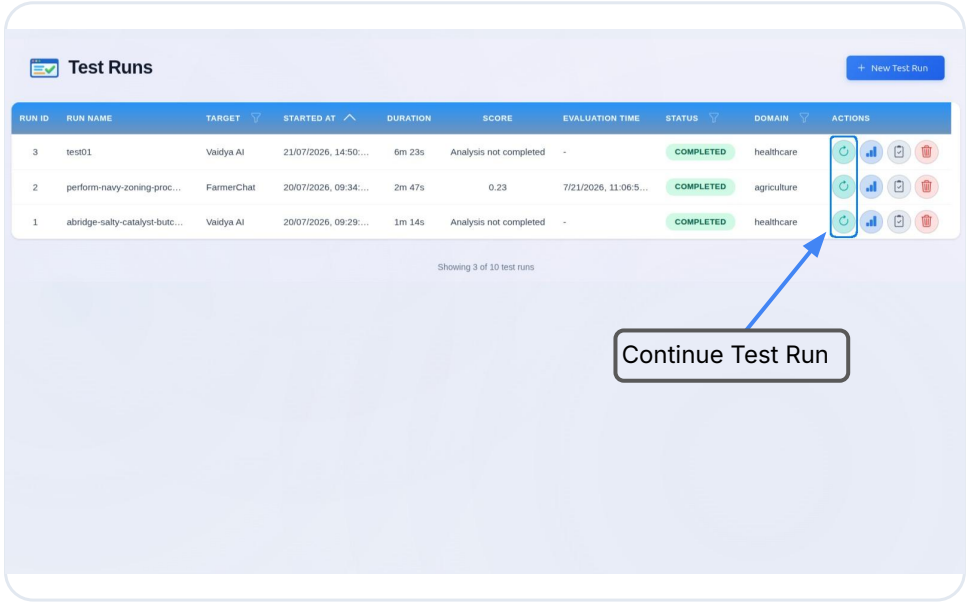
Features a dedicated continuous execution control (**cyan refresh icon**) to resume incomplete runs, re-execute existing configurations, or evaluate newly added test cases.

Integrated Action Controls

Provides one-click controls for **Re-run/Continue** via Interface Manager, **Analytics/Timeline** to view execution dashboards, and **Evaluation Reports** to export detailed results.

Domain-wise Filtering

Filter execution history by target domain (e.g., **Healthcare, Agriculture**) or status for fast auditability and quick navigation across historical datasets.



Automated Multi-Metric Evaluation and Reasoning Engine

Response Analysis & Scoring

Automated Evaluation Trigger

Initiated post-execution, the analysis engine systematically evaluates every collected target response against its designated **ground truth**, user prompts, and system guardrails.

Strategy-Driven Scoring

Dynamically applies specific evaluation strategies (such as semantic similarity or **LLM-as-a-judge**) tailored to the exact metric assigned to each test case.

Live Execution Loop

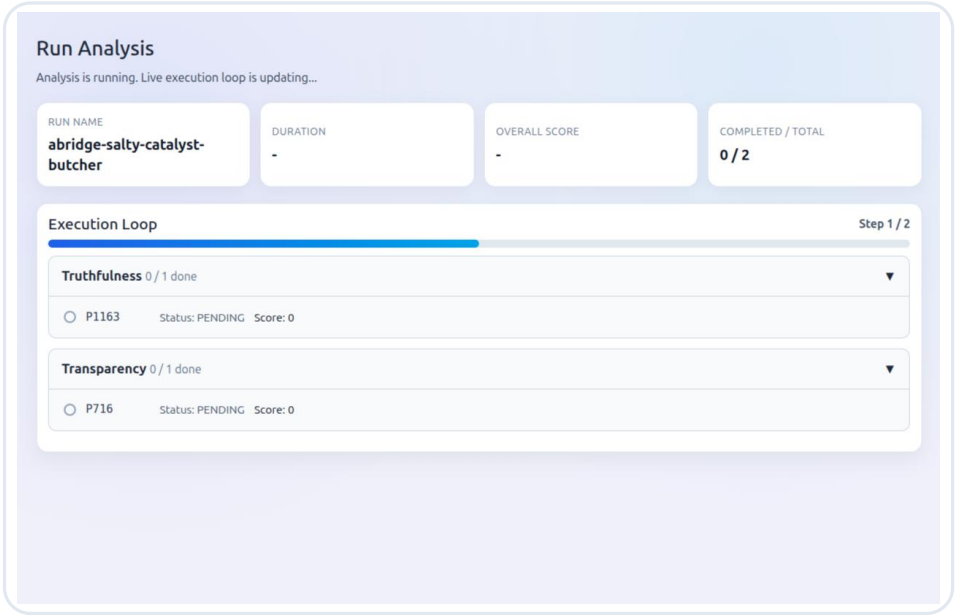
Features a real-time tracking interface that visually monitors progress, displaying **live status**, duration, and completed vs. total items.

Granular Metric Breakdown

Organizes ongoing analysis by metrics, allowing users to expand accordions to monitor specific test cases (**P1163, P716**) and view real-time scores.

Explainable Reasoning

Beyond numerical scores, the engine attaches detailed, **explainable reasoning** to ensure full transparency in the AI assessment process.



Categorizing Scoring Logic for Comprehensive Assessment

Strategy-Based Design: Evaluation Paradigms

Rule-Based Strategies

Deterministic algorithms utilizing exact text matches, regex, and structural validation. Highly efficient and ideal for evaluating explicit constraints like format adherence or strict keyword inclusion.

LLM-as-a-Judge Strategies

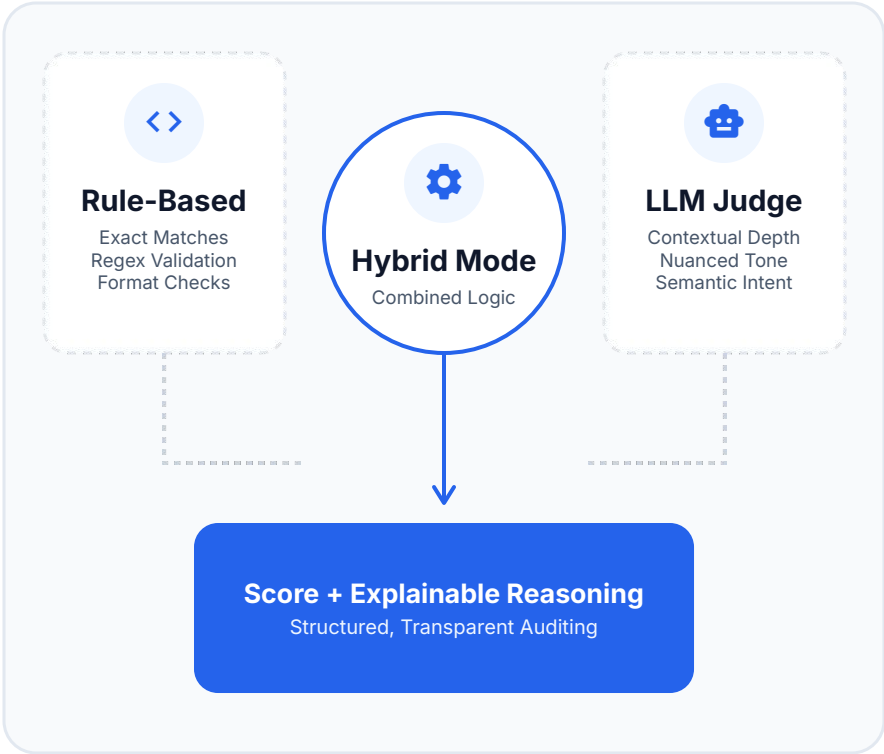
Leverages specialized **secondary LLMs (evaluators)** to conduct nuanced, context-aware semantic assessments. Excels at scoring qualitative and subjective metrics like coherence and empathy.

Hybrid (Mixed) Approaches

Combines the **speed of deterministic rules** with the deep semantic understanding of LLM evaluators to ensure both structural validity and qualitative tone appropriateness.

Dynamic Metric Alignment

The framework seamlessly routes target responses to the most **appropriate strategy type** based on the specific metric defined in the test plan, optimizing tool selection.



Comprehensive Insights and Explainable Reporting

Report Generation

Instant PDF Export

With a single click on the report icon from the Test Runs dashboard, the system dynamically compiles and exports a detailed, multi-page PDF summary.

Experiment Overview

The report header provides essential metadata, clearly documenting the Target Name (e.g., Llama3.2:1b), Run Name, Execution Timestamp, and the Total number of Test Cases.

Target Evaluation Run Summary

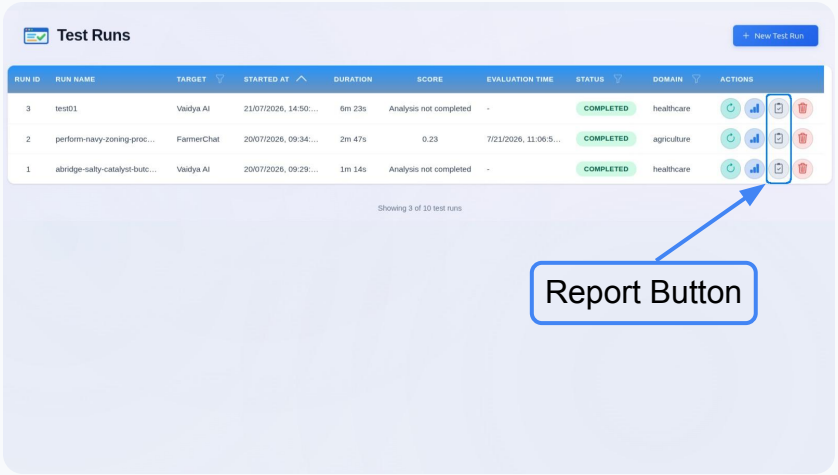
Features an auto-generated executive summary that synthesizes the model's overall performance, highlighting high-level strengths and systemic weaknesses.

Detailed Scores Table

Organizes performance data clearly by Plan Name, Metric Name, and numerical Score (0-1), providing a structured view of the evaluation.

Granular Metric & Plan Summaries

For every score, the report includes a detailed "Metric Summary" offering specific, explainable reasoning alongside a broader "Plan Summary".



Interactive Dashboard Integration

Seamlessly launch and view comprehensive PDF report exports directly from the Test Runs table UI.

Explore the CeRAI Conversational AI Evaluation Platform Live

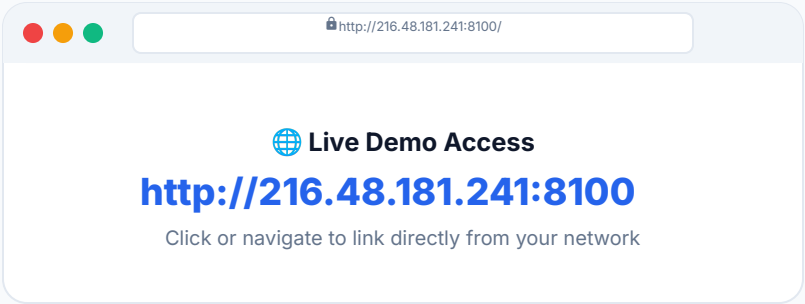
Interactive Demo

Live TDMS Exploration
Experience hands-on management of 10K+ curated test cases, domain taxonomies, and multi-platform target applications.

Real-Time Test Orchestration
Trigger, configure, and monitor live evaluation runs using the Test Case Execution Dashboard (TCED).

Multi-Strategy Analytics
Observe rule-based, LLM-as-a-judge, and hybrid strategies assessing live model responses across 7 core dimensions.

Automated Report Generation
View live execution timelines and download full PDF evaluation reports with explainable reasoning.



Scan for Mobile Access

Scan this QR code with your mobile or tablet camera to experience the live platform instantly.

iOS & Android Compatible

Phase 03

Deployment & Demonstration

Package and deploy the system architecture to a secure environment, verifying performance, request routing, and database integrations through live demonstrations.



The Deployment Dilemma

Taming the Complexity of a Multi-Service Architecture

01

The Multi-Service Challenge

The platform is not a single script, but a highly complex ecosystem of interconnected modules, including evaluation engines, data services, and databases.

02

"Works on My Machine" Syndrome

Managing intricate dependencies, varying OS environments, browser drivers, and runtime versions makes manual local setups highly error-prone.

03

The Distribution Hurdle

Sharing the platform becomes a massive operational burden when onboarding requires external users to follow pages of complex, manual installation guides.

"How do we package this entire, complex web of services into a single, standardized unit that anyone can launch instantly?"

Solution on the further slide...

Encapsulating Complex Microservices for Reproducible Execution

Containerized Deployment with Docker

Microservice Orchestration: Packages over 10 distinct system components—including multiple backends, databases, and AI models—into a single, unified docker-compose stack.

Automated Dependency Handling: Enforces strict startup rules and continuous health checks to eliminate container race conditions.

Overcoming UI Automation Blindspots: Exposes a native VNC interface to visually monitor and debug headless browser execution.

Native Hardware Integration: Directly allocates host GPU resources to containers to drive high-performance AI inference.

Persistent State & Artifacts: Utilizes mapped volumes to securely persist databases, execution logs, and model weights across restarts.



View Architecture Diagram →
See next page for the detailed system topology & routing

Mitigating Multi-Port Vulnerabilities and Centralizing Routing

NGINX Reverse Proxy: Securing the Perimeter

Security Concerns

Brute Force Targets

Multiple open ports create more entry points for scanners.

The Multi-Port Vulnerability

Exposing 5+ distinct microservice ports directly to the public internet creates a massive attack surface.

API Exposure Risks

Frameworks (like FastAPI) generate interactive docs that provide blueprints of internal APIs if left exposed. Unprotected API documentation revealing internal schemas.

Securing The Perimeter

Single Secure Entry Point

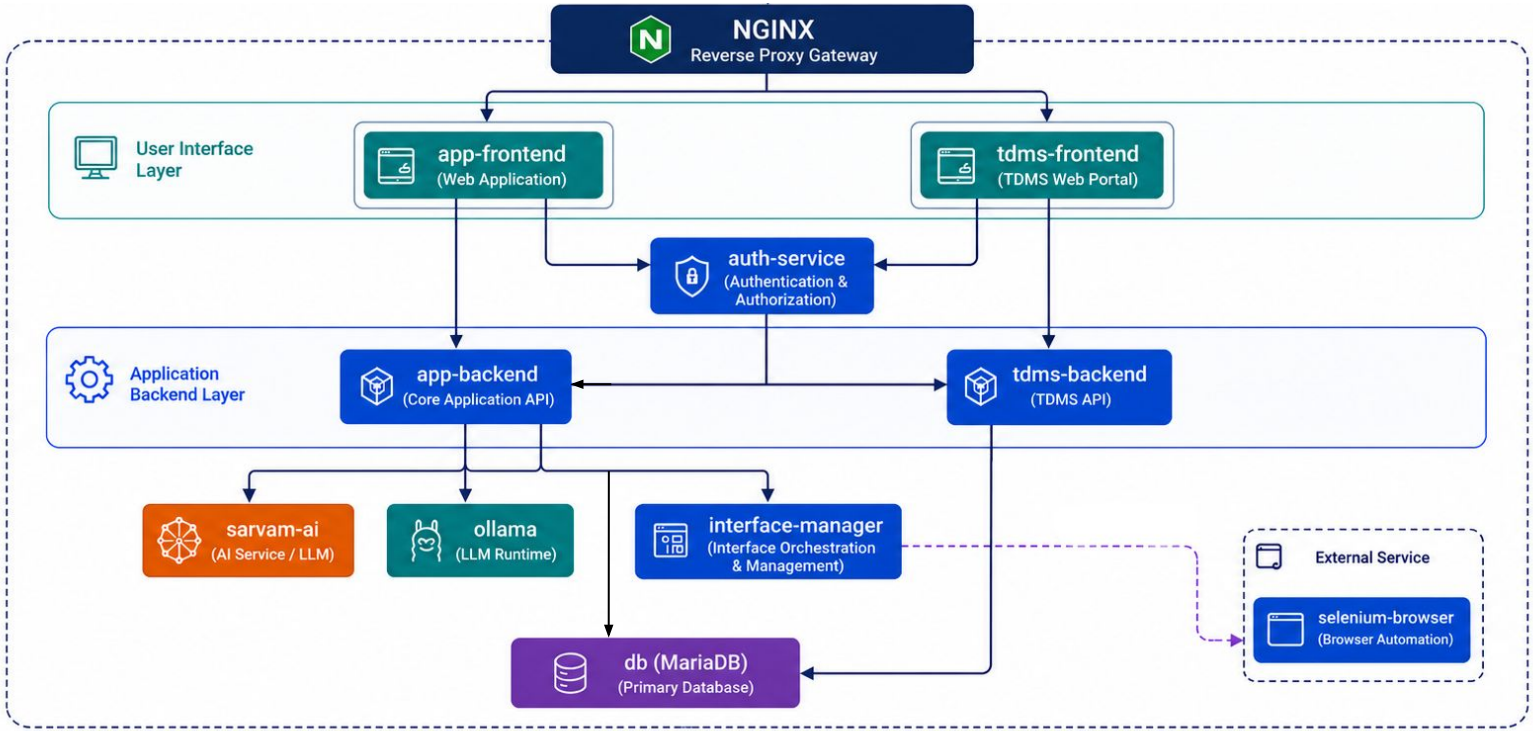
NGINX intercepts all incoming traffic on Port 80/443 and internally routes it to the correct, isolated Docker containers.

Advanced Rules & Protocol Support

- **Security Blocks:** Explicitly returns 403 Forbidden for documentation endpoints to shield the internal API.
- **WebSocket & VNC:** Seamlessly maintains persistent connections critical for live streaming traffic.

Interactive Service Dependency & Routing Map

System Architecture Topology



Project Resources

Thank you for your time

GitHub Repository

AI Evaluation Tool Documentation Web App

Sample AI Evaluation Report

Systematic Discovery of Risks in Conversational AI Systems

Trustworthy & Reliable Conversational AI through Automated Red Teaming

Dr. Sudarsun Santhiappan

Centre for Responsible AI (CeRAI)
Indian Institute of Technology Madras

Objective

What we aim to achieve

To design an automated evaluation platform for testing conversational AI systems that integrates structured test plans, metrics, and strategies, enabling standardized model & system assessment across tasks, domains, and languages.